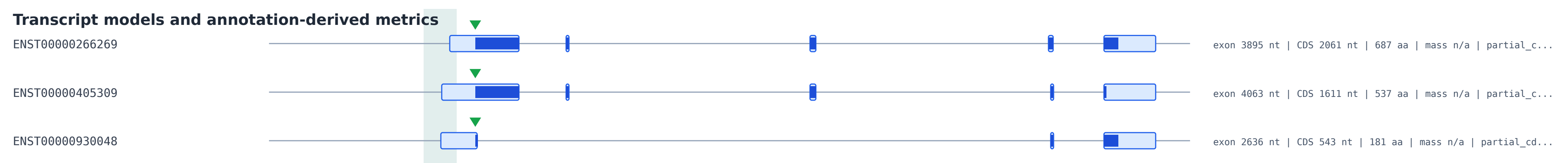
PATZ1 promoter-reporter architectures

5' -> 3' display, genomic 31351605 -> 31324804 | strand - | assembly GRCh38 | annotation Ensembl 116 prepared GRCh38; selected reporter transcripts | upstream 5000 bp | downstream 1000 bp

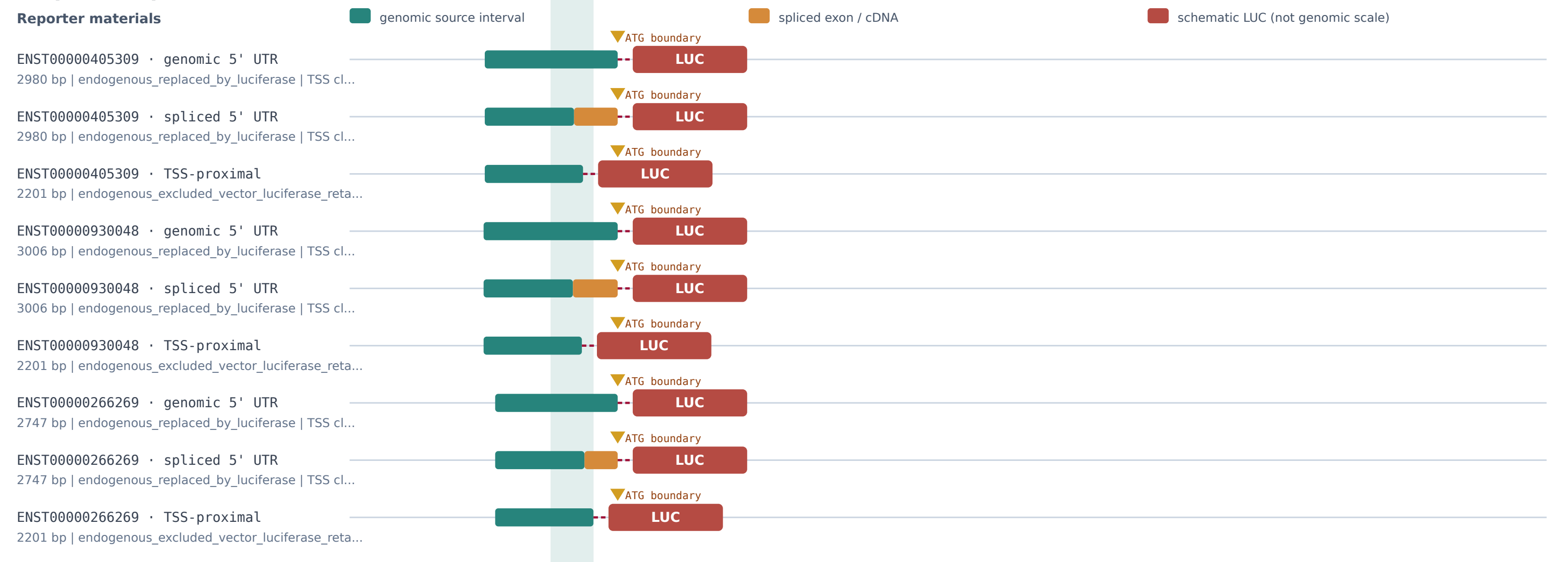
Canonical reporter comparison: 2 TSS class(es), 3 transcript(s), 9 explicit architecture row(s); normalized evidence remains observational/predictive context.

TSS stretches: reference for similarity below

Transcript models and annotation-derived metrics

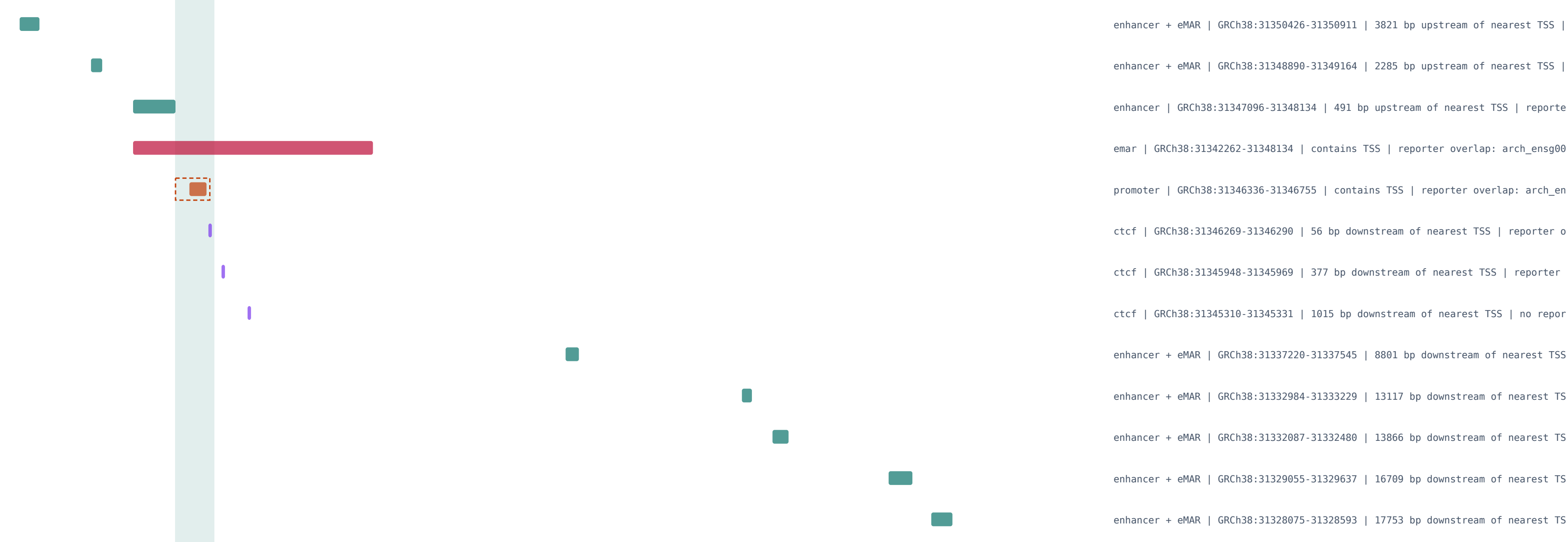

ENST00000266269
exon 3895 nt | CDS 2061 nt | 687 aa | mass n/a | partial_c...
ENST00000405309
exon 4063 nt | CDS 1611 nt | 537 aa | mass n/a | partial_c...
ENST00000930048
exon 2636 nt | CDS 543 nt | 181 aa | mass n/a | partial_cd...

Proposed reporter architectures


source intervals use the shared axis; synthetic tails are schematic
Reporter materials
genomic source interval
spliced exon / cDNA
schematic LUC (not genomic scale)
ATG boundary
LUC
ENST00000405309 · genomic 5' UTR
2980 bp | endogenous_replaced_by_luciferase | TSS cl...
ENST00000405309 · spliced 5' UTR
2980 bp | endogenous_replaced_by_luciferase | TSS cl...
ENST00000405309 · TSS-proximal
2201 bp | endogenous_excluded_vector_luciferase_reta...
ENST00000930048 · genomic 5' UTR
3006 bp | endogenous_replaced_by_luciferase | TSS cl...
ENST00000930048 · spliced 5' UTR
3006 bp | endogenous_replaced_by_luciferase | TSS cl...
ENST00000930048 · TSS-proximal
2201 bp | endogenous_excluded_vector_luciferase_reta...
ENST00000266269 · genomic 5' UTR
2747 bp | endogenous_replaced_by_luciferase | TSS cl...
ENST00000266269 · spliced 5' UTR
2747 bp | endogenous_replaced_by_luciferase | TSS cl...
ENST00000266269 · TSS-proximal
2201 bp | endogenous_excluded_vector_luciferase_reta...

Ensembl-annotated regulatory regions

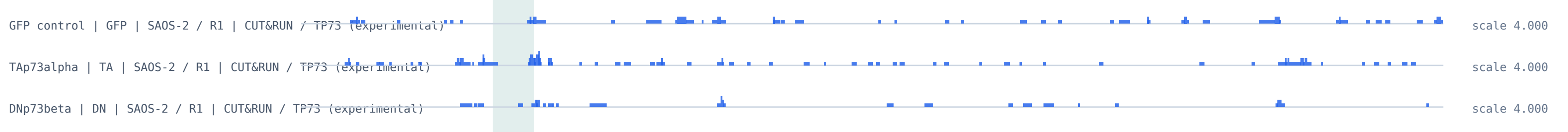
Ensembl Regulation | release 2026-08 | pipeline 2.1 | GRCh38



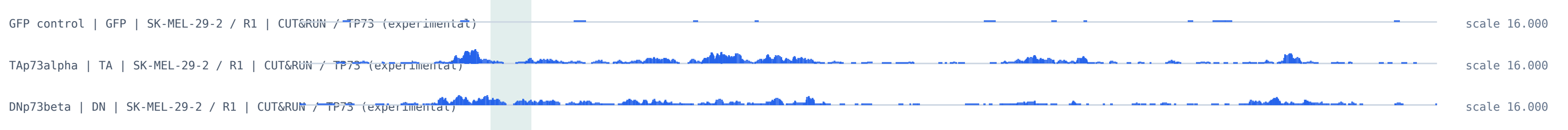
7 exact-geometry eMAR row(s) combined with matching displayed features; complete rows remain in JSON.

Projected occupancy grouped by declared biological context

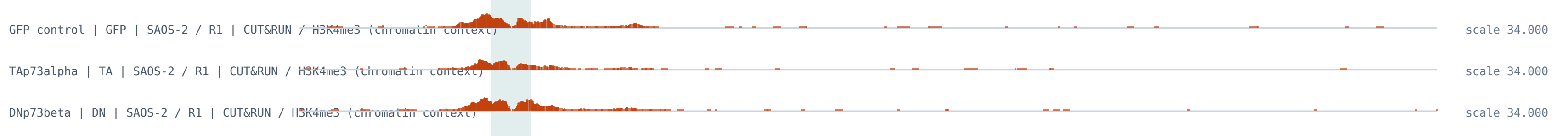
TP73 occupancy — SAOS-2 | shared within group | observed group max 4.000



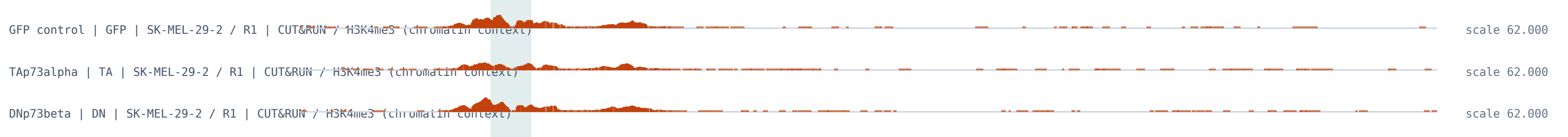
TP73 occupancy — SK-MEL-29-2 | shared within group | observed group max 16.000



H3K4me3 chromatin context — SAOS-2 | shared within group | observed group max 34.000



H3K4me3 chromatin context — SK-MEL-29-2 | shared within group | observed group max 62.000



Predicted TF binding scores

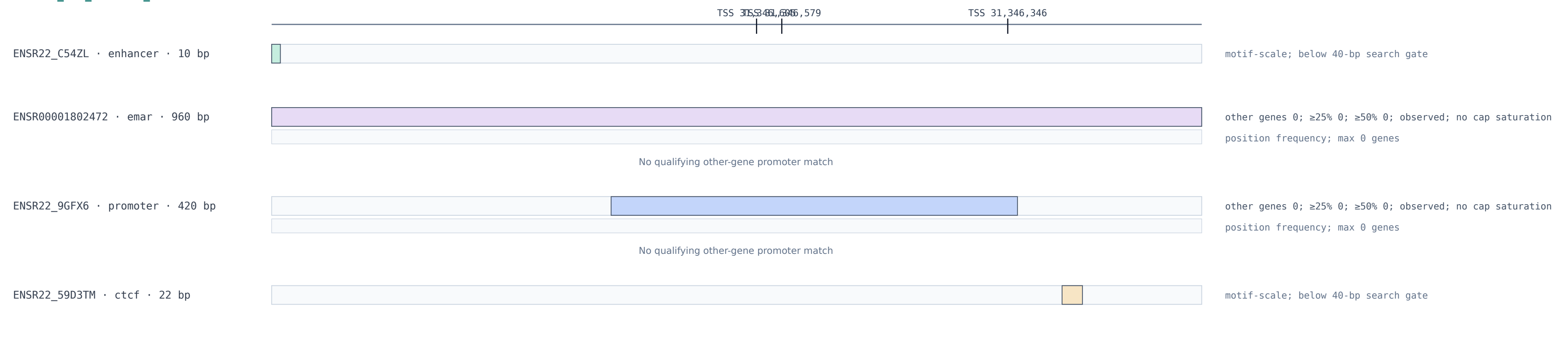
TP53 JASPAR PWM | jaspar_pwm | TP53 | MA0106.3 | llr_background_tail_log10
-log10 Ptail
F solid | R dashed
display 0.000..4.994 | -log10 modeled background-tail probability | matrix_spe
JASPAR 2026 CORE matrix-specific PWM score; not calibrated as biochemical affi
TP63 JASPAR PWM | jaspar_pwm | TP63 | MA0525.2 | llr_background_tail_log10
display 0.000..5.455 | -log10 modeled background-tail probability | matrix_spe
TP73 JASPAR PWM | jaspar_pwm | TP73 | MA0861.2 | llr_background_tail_log10
display 0.000..5.538 | -log10 modeled background-tail probability | matrix_spe
SP1 JASPAR PWM | jaspar_pwm | SP1 | MA0079.5 | llr_background_tail_log10
CGGBP1 JASPAR PWM | jaspar_pwm | CGGBP1 | MA2538.1 | llr_background_tail_log10
HELT JASPAR PWM | jaspar_pwm | HELT | MA2628.1 | llr_background_tail_log10
IRF9 JASPAR PWM | jaspar_pwm | IRF9 | MA0653.1 | llr_background_tail_log10
PROX1 JASPAR PWM | jaspar_pwm | PROX1 | MA0794.1 | llr_background_tail_log10
FOXF2 JASPAR PWM | jaspar_pwm | FOXF2 | MA0030.2 | llr_background_tail_log10
MYBL2 JASPAR PWM | jaspar_pwm | MYBL2 | MA0777.1 | llr_background_tail_log10
PLAG1 JASPAR PWM | jaspar_pwm | PLAG1 | MA0163.1 | llr_background_tail_log10
STAT1 JASPAR PWM | jaspar_pwm | STAT1 | MA0137.4 | llr_background_tail_log10
PATZ1 JASPAR PWM | jaspar_pwm | PATZ1 | MA1961.2 | llr_background_tail_log10
ELK4 JASPAR PWM | jaspar_pwm | ELK4 | MA0076.3 | llr_background_tail_log10
REST JASPAR PWM | jaspar_pwm | REST | MA0138.3 | llr_background_tail_log10
LMX1B JASPAR PWM | jaspar_pwm | LMX1B | MA0703.3 | llr_background_tail_log10
POU2F2 JASPAR PWM | jaspar_pwm | POU2F2 | MA0507.3 | llr_background_tail_log10
TFAP2C JASPAR PWM (MA0524.3) | jaspar_pwm | TFAP2C | MA0815.1 | llr_background_tail_log10
TFAP2C JASPAR PWM (MA0814.3) | jaspar_pwm | TFAP2C | MA0815.1 | llr_background_tail_log10
TFAP2C JASPAR PWM (MA0815.1) | jaspar_pwm | TFAP2C | MA0815.1 | llr_background_tail_log10
ZBED1 JASPAR PWM | jaspar_pwm | ZBED1 | MA0749.2 | llr_background_tail_log10
TGIF2 JASPAR PWM | jaspar_pwm | TGIF2 | MA0797.1 | llr_background_tail_log10
GATA5 JASPAR PWM | jaspar_pwm | GATA5 | MA0766.3 | llr_background_tail_log10
HES7 JASPAR PWM | jaspar_pwm | HES7 | MA0822.1 | llr_background_tail_log10
KLF15 JASPAR PWM | jaspar_pwm | KLF15 | MA1513.2 | llr_background_tail_log10
NFKB2 JASPAR PWM | jaspar_pwm | NFKB2 | MA0778.2 | llr_background_tail_log10
POU2F1 JASPAR PWM | jaspar_pwm | POU2F1 | MA0785.2 | llr_background_tail_log10
SATB1 JASPAR PWM | jaspar_pwm | SATB1 | MA1963.2 | llr_background_tail_log10
IRF2 JASPAR PWM | jaspar_pwm | IRF2 | MA0051.2 | llr_background_tail_log10
MEF2B JASPAR PWM | jaspar_pwm | MEF2B | MA0660.1 | llr_background_tail_log10

TSS-local Ensembl-feature promoterome similarity

Transcript-oriented −500/+200 bp; features clipped to the displayed stretch; BLASTN ≥40 bp, ≥80% identity, E≤1e-05; self-locus and same-gene targets excluded.

Blue frequency strips count distinct other genes by query position. Numbered blocks give target-promoter order; red outlines mark order/orientation breaks.

PATZ1_tss_stretch_1



Similarity is structural evidence; Ensembl classes, predicted TFBS and CUT&RUN enrichment do not prove reporter activity or sufficiency.

Reporter interpretation boundaries

Transcript annotation alone does not establish which TSS is used in the assayed cells.
CUT&RUN overlap is occupancy/enrichment evidence, not proof of TSS usage, promoter activity, direct regulation, biochemical affinity, or luciferase output.
Reporter geometry does not verify the identity of a physical laboratory construct.

Warnings / interpretation checks

Evidence provenance

ensembl_regulation_annotation_index: ensembl_regulation_2026_08_grch38
ensembl_regulation_annotation_intervals: ensembl_regulation_2026_08_grch38
isoform_panel: PATZ1_panel
occupancy_layout: tp73_saos2,tp73_skmel29_2,h3k4me3_saos2,h3k4me3_skmel29_2
predicted_tf_score_jaspar_pwm: local_jaspar_registry
projected_occupancy_track: E-MTAB-15709:h3k4me3:saos2:DN:R1
projected_occupancy_track: E-MTAB-15709:h3k4me3:saos2:GFP:R1
projected_occupancy_track: E-MTAB-15709:h3k4me3:saos2:TA:R1
projected_occupancy_track: E-MTAB-15709:h3k4me3:skmel29-2:DN:R1
projected_occupancy_track: E-MTAB-15709:h3k4me3:skmel29-2:GFP:R1